1. Excretion in Humans

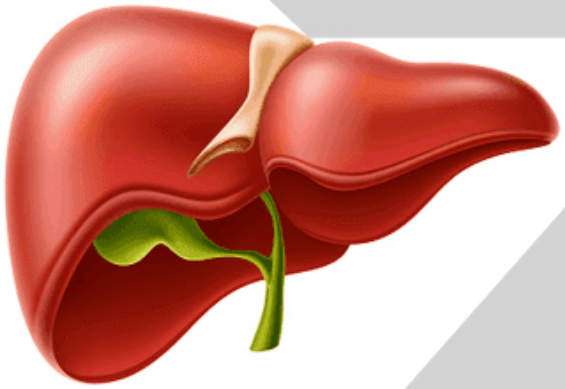
1.1. Excretion

Excretion: the removal of toxic materials from organisms, the waste products of metabolism (chemical reactions in cells including respiration) and substances in excess of requirements.

- Substances should include carbon dioxide (lungs), urea, excess water and ions (kidney).
- The importance of excretion is due to the **toxicity of the urea**.

1.2. Function of Liver

The role of the liver is in the **assimilation** of amino acids by converting them to proteins.



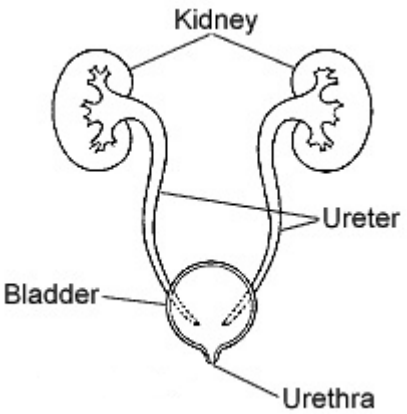
Deamination: A critical process where the removal of an amino group (-NH2) from a molecule, usually an amino acid. This reaction mostly takes place in the liver and is an essential part of the body's metabolism of proteins and nitrogen.

Assimilation: uptake and use of nutrients by body cells

- Urea is formed in the liver from excess amino acids.
- Alcohol, drugs & hormones are broken down in the liver.

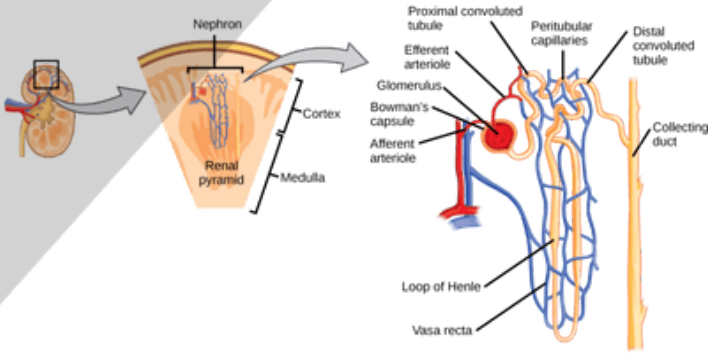
1.3. Function of Kidney

- Removal of urea and excess water and the re-absorption of glucose and some salts



- **Cortex:** contains Bowman's capsules and coiled tubules.
- **Ureter:** carries urine from the kidney to the bladder
- **Medulla:** has loops of Henlé and collecting ducts
- **Urethra:** carrying urine from the bladder to the outside.
- **Bladder:** stores urine
- **Renal artery:** brings wastes and water from the blood
- **Renal vein:** reabsorbs water and functional molecules and leaves wastes behind

1.4. Structure and Function of the Nephron



1. **Ultrafiltration:**
   - blood from the renal artery enters the glomerulus under high pressure.
   - Water, urea, salts, and glucose are forced into the Bowman's capsule.
   - Blood cells and large proteins cannot pass through.
2. **Selective reabsorption:** in the tubule, two-thirds of the salt and water and all the glucose move out of the nephron by active transport. These substances are reabsorbed back into the blood capillary.
3. **Loop of Henlé:** this part is permeable to water but not salt. Water is drawn out of the filtrate in the nephron by osmosis because of the low water potential of the medulla tissue fluid.
4. **Collecting duct:** the remaining substances move into the collecting duct, forming urine.

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